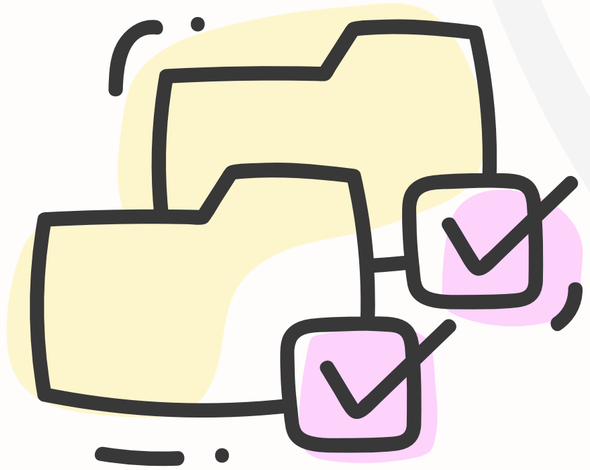


Data Analysis, Statistical Tests, and Applied Computer Science

NextInResearch

Deborah Torrico-Pardo & Dev Srivastava

Check-In



Take one minute to individually or with your group, do a self-assessment with where you are in your project

3 Ways to Study the World:

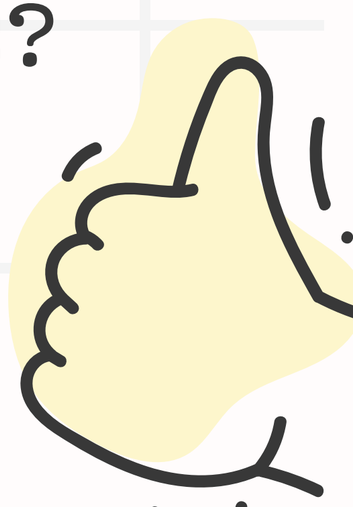
- **Quantitative Analysis:** Uses numbers and math to test theories. (Example: Tracking average test scores.)
- **Qualitative Analysis:** Uses words, audio, and video to understand meanings and behaviors. (Example: Interviewing people about their feelings.)

Mixed Methods: Combining both!



TURN AND TALK

Discussion Question: "If you and wanted to know why students like or dislike the cafeteria food, would you use Quantitative (numbers/surveys) or Qualitative (talking/interviews) methods? Why?



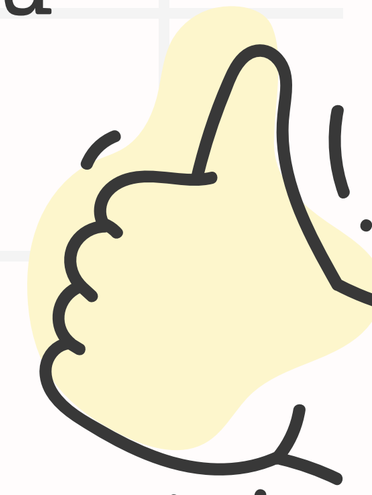
Steps for Data Analysis

Step 1: Data Preparation

- **Data Cleaning:** Fix inaccurate or irrelevant records
  - **Ex:** a participant accidentally types "290" instead of "29" for age
- **Data Transformation:** Convert data from one format into another to make it ready for analysis tools (for QUALITATIVE data)
  - **Ex:** Changing gender ("M", "male", "Male") into a standardized numeric code (1 for Male, 2 for Female)
- **Handling Missing Data:** Delete or adjust any missing data

TURN AND TALK

Discussion Question: Person A, B, C, and D filled out a survey. Person A, C, and D's ages are 14, 18, and 15, respectively. Person B's AGE is missing. What can you do with Person B's data?



TURN AND TALK

Mini-Lesson: You can either delete person B from the dataset, or IMPUTE data, or substitute null/missing values with estimates!

Ex: since the other ages in the survey were 14, 18, and 15, you can fill in Person B's data with the average, 16.

Step 2: Analytical Approaches (Quantitative or Qualitative?)

A: Quantitative

Uses mathematical frameworks to find patterns, averages, and causal relationships. Consists of **TWO** main phases

1. Descriptive Statistics

Definition: Describing the basic features of a dataset. Doesn't look for deep relationships, just a summary

Key Metrics: Mean (average), Median (middle value), Mode (most common value), and Standard Deviation (how spread out the numbers are).

Example: An education researcher calculates that the mean test score of a classroom is 85%, with a standard deviation of 5 points, meaning most students scored very close to the average.

A: Quantitative

2. Inferential Statistics

Definition: Taking data from a sample and making broader predictions or generalizations about a larger population.

Key Techniques:

Regression Analysis: Testing how much an independent variable changes a dependent variable.

T-Test / ANOVA: Comparing the averages of different groups to see if they are statistically significant or just different by luck.

Important terms:

p-Value: A p-value is the probability your results occured by random chance alone. A p-value under 0.05 means something is happening, and therefore significant

TURN AND TALK

Discussion Question: If you flipped a coin and landed on heads, what is the EMPIRICAL (by experiment) probability of heads? What if you flipped the coin 1000 times?

B: Qualitative

Examples:

- **Data Coding:** Turning a text response into a label to categorize them
  - ***Example*:** In a study on job satisfaction, whenever an interviewee mentions "exhaustion" the researcher labels that text with the code [Burnout].
- **Content Analysis:** Quantifying the presence of certain words, concepts, or themes within text to determine their frequency and context
  - ***Example*:** Counting how many times the word "safety" is used in an airline's internal incident reports to measure their safety focus.

Step 3: Interpretation & Visualization!

Step 3: Interpretation & Visualization

The final step is translating the math or the themes into answers that directly address the original research question.

- **Data Visualization:** Presenting data visually so human eyes can immediately spot trends
  - Example: Using a scatter plot with a trendline to show a clear positive correlation between hours spent studying and final exam grades.
- **Contextualization:** Explaining why the numbers matter while explicitly naming any limitations of the study. Also consider APPLICATIONS of your results

Applied Computer Science Mini-Lesson

TURN AND TALK

Discussion Question: What is the definition of Artificial Intelligence? What comes to mind when you hear that word? What are ways you've used AI in the past?

How can you use Applied Computer Science (responsibly) in your research project:

- **Analyzing data:** If you want to visualize data, code an app, or automate a process, you can write code to do so.
- **Key skills to learn for your own projects/high school:** Coding in python! Often, introductory courses explain a lot about what computers are and how they work. Python is a universal language and it's a great way to start and once you learn the fundamentals a lot can be built off it.
- **ChatGPT, Claude, Gemini:** What are large language models, brief introduction (We will do the rest in the next workshop).





TURN AND TALK

Discussion Question: What is one way you can apply computer science in your own research project?

FUN ACTIVITY!

Activity: Are skittles different flavors?

Instructions:

- Pair up with someone around you! We will hand each of you a bag of skittles.
- Take turns closing your eyes and taste testing each skittle and then guessing what color skittle it is. Have your partner write down whether you correctly or incorrectly guessed.
- Repeat with all skittles in the bag
- Determine whether you thought your results were significant (ex: you guessed all correct!) or pure luck (you only got one correct)

Thank You!